

MultiSticker: Intent-Guided Multimodal Sticker Retrieval with Dialogue Memory

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Abstract

*Stickers are widely used in messaging platforms to express affect, humor and intent. Retrieving the right sticker for a dialogue turn is challenging because (i) the candidate bank is large and dominated by animated formats, (ii) the correct reply depends on long-range conversational state rather than just the last turn, and (iii) each context typically has only one observed positive even though many semantically equivalent stickers exist. We present **MultiSticker**, an intent-guided sticker retrieval system that combines recent dialogue context, retrieved long-term session memory, and an LLM-generated reply intent with a frozen-or-adapted OpenCLIP sticker bank. The paper is organized around two experiments. First, we test whether dialogue memory is useful by comparing frozen CLIP and a learned retrieval head with and without retrieved memory. Frozen CLIP already retrieves semantically related stickers ($\text{Group@10} = 0.9078$), but fails at exact sticker selection ($\text{Exact@30} = 0.0028$); adding a learned head and retrieved memory raises Exact@30 to 0.1263. Second, we ask which parameter-efficient SFT direction is best: text-side LoRA, image-side LoRA, or dual LoRA. On the completed medium-scale U-Sticker run, dual LoRA gives the best trained-model semantic Group@5/10 performance, while image-side LoRA gives the strongest exact top-5 ranking. We argue that for a chat sticker panel, semantic Group@5/10 is the right north-star metric and exact retrieval should be treated as a secondary within-group ranking diagnostic.*

1. Introduction

When users reply with stickers in a chat, the choice depends on local context (what was just said), long-range state (who is talking, prior tone, recurring topics) and the communicative *intent* behind the reply (e.g. gratitude, encouragement, teasing). A retrieval system that only looks at the last few turns frequently selects stickers with the wrong affect or the wrong recurring style for a given user stream.

Conversely, a system that uses only image-side similarity ignores the dialogue entirely.

We focus on the U-Sticker corpus, which contains real multilingual chats spanning crypto, gaming and social-domain channels. Several properties make sticker retrieval different from generic image-text retrieval. First, the candidate set is sparse and animated: 42,319 decodable stickers across .png, .gif and .webm. Second, supervision is *single-positive*: each turn records the one sticker the user actually sent, even though many other stickers in the bank are equally valid. Third, turns frequently contain code-switching and chat slang outside CLIP’s pre-training distribution.

We build on the IGSR-style design [7] of using a CLIP backbone plus a small learned head, and extend it in three directions:

- A retrieved *long-term memory* stream from previous sessions of the same user/file, encoded by multilingual E5 [8].
- An LLM-generated *reply intent* stream that abstracts away from the exact sticker id and describes the communicative function of the reply.
- A unified pipeline that handles .png, .gif and .webm via per-frame OpenCLIP encoding and mean pooling, and supports parameter-efficient adaptation through LoRA [2] on the text side, image side, or both.

Contributions. (1) A memory ablation showing that retrieved long-term dialogue memory mainly improves fine-grained sticker-id selection, not only coarse semantic matching. (2) A modality-SFT study comparing text-side, image-side, and dual LoRA on the same all-media sticker bank, showing that dual LoRA is most balanced for semantic panel retrieval while image LoRA is best for exact top-5 ranking. (3) A reusable, self-contained MultiSticker codebase including data manifest construction, memory retrieval, OpenCLIP wrapping, multi-frame decoding, LoRA-capable training modes, and SLURM-friendly run scripts.

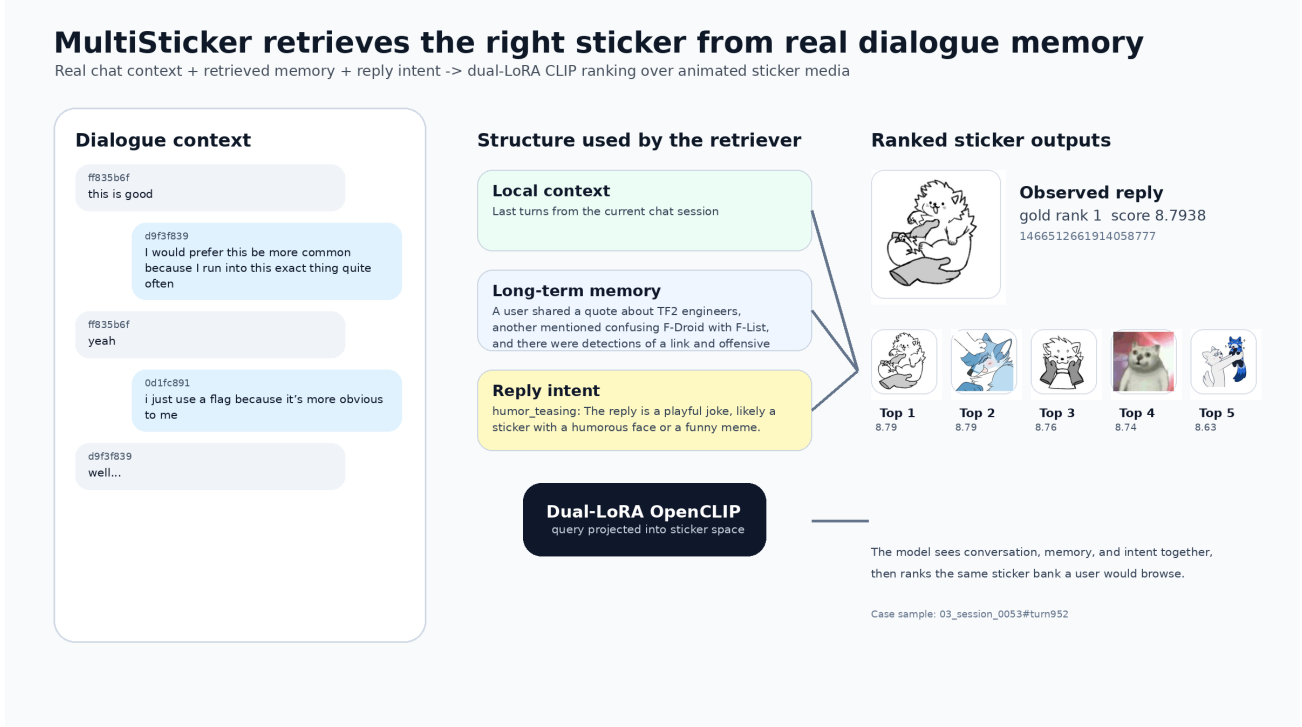


Figure 1. **MultiSticker at a glance.** A real validation dialogue is converted into three retrieval signals: local dialogue context, retrieved long-term memory, and an LLM-generated reply intent. The dual-LoRA OpenCLIP retriever projects these signals into the sticker embedding space and returns a ranked panel of real sticker candidates; animated `.webm` and `.gif` media are shown by their first frame.

2. Related Work

Vision-language retrieval. CLIP [6] popularised dual-encoder image-text retrieval, and OpenCLIP [3] provides open-weight reproductions trained on LAION-class corpora. Our system uses ViT-B/32 with the `laion2b_s34b_b79k` weights as the backbone for both the text and image branches.

Sticker / meme retrieval. Sticker reply suggestion has been studied as a multimodal retrieval and ranking problem [4, 1]. Intent-guided sticker retrieval (IGSR) [10] adds an explicit intent classification head and a group prior. We adopt this template and add long-term memory and parameter-efficient adaptation.

Memory-augmented dialogue. Long-term dialogue memory has been used to condition open-domain chatbots [9]. We use it specifically to construct a third query stream for retrieval rather than to generate text. Our memory encoder is multilingual E5 [8], chosen for code-switched chat data.

Parameter-efficient fine-tuning. LoRA [2] adds low-rank update matrices to selected linear layers, leaving the original weights frozen. We inject LoRA into OpenCLIP’s `out_proj`, `c_fc` and `c_proj` modules through PEFT [5], and isolate text-side, image-side and dual adaptation through name-based parameter masking.

3. Method

3.1. Pipeline overview

For each turn t where the user sent a sticker s_t^* , we form three text inputs: the immediate *context* c_t (up to 12 prior turns), the retrieved *memory* m_t (concatenation of the top-3 prior session summaries), and the LLM *intent* i_t (a short description of what the sticker should express). These are encoded independently by the OpenCLIP text encoder f_T . Sticker images s are encoded by the OpenCLIP image encoder f_I (with mean-pooling over frames for animated formats). The trainable head g_θ projects the concatenation into the image space:

$$\mathbf{q}_t = \text{norm}\left(g_\theta([f_T(c_t); f_T(m_t); f_T(i_t)])\right). \quad (1)$$

Retrieval logits are $\mathbf{z}_t = \mathbf{q}_t \mathbf{B}^\top / \tau$, where $\mathbf{B} \in \mathbb{R}^{N \times d}$ is the (optionally adapted) OpenCLIP image bank and τ is a learned temperature. A second head predicts the sticker’s intent group y_t , and we minimise

$$\mathcal{L} = \text{CE}(\mathbf{z}_t, s_t^*) + \lambda \text{CE}(\mathbf{z}_t^{\text{int}}, y_t), \quad \lambda=0.2. \quad (2)$$

3.2. Long-term memory retrieval

Each raw chat file is split into sessions by a 12-hour inactivity gap. For sample t , we collect up to 8 prior sessions from the same file. Their LLM-generated session



Figure 2. End-to-end MultiSticker pipeline. Raw U-Sticker JSON is split into 12-hour sessions; an LLM produces a per-session memory and a per-sample intent label/text; multilingual E5 retrieves the top-3 historical sessions for the current turn; OpenCLIP independently encodes the context, memory and intent strings; the `IntentGuidedRetriever` head projects their concatenation into the CLIP image space and ranks the sticker bank by cosine similarity. An auxiliary intent classifier predicts the sticker’s group and supplies a prior for the two-stage rerank.

summaries are encoded with the `E5 passage:` prefix; the current session text is encoded with the `query:` prefix. After L2 normalisation, the dot products rank the prior sessions and we keep the top-3. The selected memory texts are concatenated and stored as m_t , then encoded by OpenCLIP at training time. Crucially, memory enters the model through the `text` encoder, not as a separate vector, which lets the retrieval head learn to reweight memory against the local context.

3.3. LLM annotation

We annotate every session with a ≤ 4 -sentence `session_memory_text` (used as the passage corpus for memory retrieval), and every sticker-reply sample with one `intent_label` from a 10-class taxonomy (celebration_approval, humor_teasing, encouragement_support, gratitude, affection_care, confusion_curiosity, frustration_disapproval, disappointment_sadness, surprise_reaction, neutral_acknowledgment) plus a free-form `intent_text`. The intent label feeds the auxiliary head and induces the semantic-group label space; the intent text becomes the third query stream.

3.4. Multi-frame sticker encoding

.png stickers are encoded as one-frame clips. .gif and .webm are decoded to all frames (FFmpeg for video, PIL for GIF), each frame is encoded by the OpenCLIP image encoder, and the per-frame embeddings are mean-pooled and renormalised. During image-LoRA training, the full bank is re-encoded once per epoch without gradients; for each batch we additionally re-encode only the unique *positive* stickers with gradients (animated positives capped at 4 frames) and patch them into a cloned bank, which keeps backprop tractable for a 42K-sticker bank.

3.5. Tuning modes and experimental axes

The code exposes two independent axes. The first is the memory strategy: `retrieved.topk` uses E5 retrieval over previous session summaries, while `disabled` replaces the memory stream with a null placeholder. This gives a clean memory ablation because the architecture, data split, and training objective remain unchanged.

The second axis is the SFT mode. **direct_clip** uses no learned head: the query is just $f_T(c_t)$ scored against the frozen image bank, giving a strong off-the-shelf VLM baseline. **head_only** trains the retrieval head on top of frozen OpenCLIP encoders. **text_lora** additionally injects LoRA

($r=8$, $\alpha=16$) into the text branch. **image_lora** injects LoRA into the visual branch only, while **dual_lora** adapts both branches. Only LoRA parameters and the head are trainable; all original OpenCLIP weights are frozen.

4. Experiments

4.1. Setup

We report the completed `rebuild_medium` setting. The manifest is constructed from 8,129 sessions and 368,503 provisional sticker-reply samples. After frequency filtering and media decoding, the experiment uses 30,000 training samples, 5,000 validation samples (4,998 after media filtering), and a 6,758-sticker bank over `.png`, `.gif`, and `.webm`. All trainable models are optimized for 5 epochs with AdamW. The retrieval head uses batch size 256; LoRA runs use batch size 96; inference batch size is 256. The CLIP backbone is OpenCLIP ViT-B/32 with `laion2b_s34b_b79k` weights. The retrieval temperature is 0.07, hidden dimension is 512, dropout is 0.1, the auxiliary intent-loss weight is 0.2, and the intent vocabulary is clustered into 64 groups. LoRA uses rank $r=8$, $\alpha=16$, dropout 0.05, LoRA LR 10^{-4} , and head LR 10^{-3} .

Compared models. We report two experiments. Experiment 1 isolates memory. `direct_clip` is the context-only off-the-shelf baseline, while `head_only` is evaluated with memory disabled and with retrieved memory. This separates what frozen CLIP already solves from the extra value of the memory-aware retriever. Experiment 2 fixes `retrieved_topk` memory and compares `head_only`, `text_lora`, `image_lora`, and `dual_lora`. This isolates whether SFT helps more on chat text, sticker imagery, or both modalities together.

4.2. Metrics

We evaluate retrieval at two granularities. The primary metric is **Group Hit@K**: a prediction is correct if any of the top- K retrieved stickers belongs to the same semantic intent group as the logged sticker. This is the most faithful proxy for a chat sticker panel, where the interface shows 5–10 plausible choices and the user mainly needs the right affect, tone, or communicative intent. We therefore treat **Group@5** and **Group@10** as the north-star metrics. As a secondary diagnostic, we also report **Exact Hit@K**, which requires recovering the single observed sticker id. Exact metrics measure within-group ranking quality, but are conservative because many unobserved stickers may be equally valid. All Hit@K metrics are higher-is-better.

4.3. Results

Table 1 answers whether memory is useful, while Table 2 and Table 3 answer which SFT modality is best. Figure 3 shows a representative `dual_lora` retrieval on a real

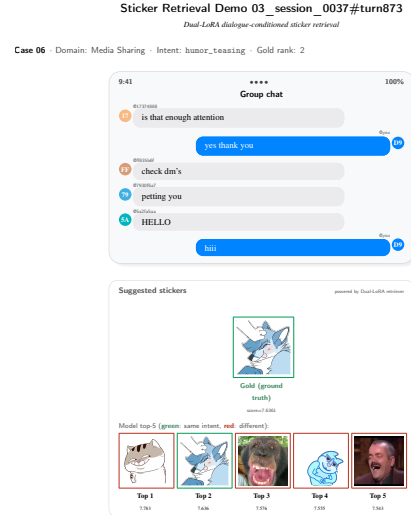


Figure 3. **Qualitative example (case 06, dual_lora).** A `humor_teasing` turn from a Media-Sharing chat. The retriever places the gold sticker at rank 2 (green frame); rank 1 is a visually similar `neutral_acknowledgment` sticker (red). All five candidates are plausible reactions for the conversation, illustrating why Group@5 is more informative than Exact@1 for sticker-panel UX. Seven additional cases (success → near-miss → failure) are listed in Appendix A.

validation turn. The memory ablation shows that off-the-shelf CLIP is already strong at coarse semantic matching, but retrieved memory is crucial for exact sticker-id selection in the learned retriever. The SFT experiment shows that `dual_lora` is the best trained model under the panel-recommendation objective: it wins overall Group@5 and is second only to the zero-shot baseline on Group@10. At the same time, `image_lora` has the strongest exact top-5 ranking, suggesting that visual adaptation is especially useful for ordering near-duplicate stickers within a semantic group.

5. Analysis and Discussion

Experiment 1: memory helps exact sticker selection.

The off-the-shelf frozen CLIP baseline is a strong and necessary comparison: with no training and no memory, it reaches Group@10 = 0.9078 and Group@30 = 0.9592. This means CLIP already understands the coarse visual-language semantics of many sticker replies. However, it does not solve the actual sticker-retrieval problem: Exact@30 is only 0.0028. The learned `head_only` model without memory improves Exact@30 to 0.0412, but adding retrieved memory raises it further to 0.1263. Group metrics move much less. The non-obvious lesson is therefore that memory is not mainly a coarse semantic booster; it helps the model choose the concrete sticker id inside an already plausible semantic neighborhood.

Model	Memory	Trainable	Group@5↑	Group@10↑	Group@30↑	Exact@5↑	Exact@30↑	MAP↑
direct_clip	disabled	none	0.8778	0.9078	0.9592	0.0012	0.0028	0.0013
head_only	disabled	head	0.8567	0.8754	0.9324	0.0082	0.0412	0.0107
head_only	retrieved	head	0.8619	0.8780	0.9406	0.0296	0.1263	0.0282

Table 1. **Experiment 1: memory ablation and off-the-shelf baseline.** Frozen CLIP is a sensible VLM baseline: it already retrieves semantically related stickers, but does not solve exact sticker-id selection. The causal memory ablation is `head_only` disabled versus retrieved. With the same head-only architecture, retrieved memory gives a large gain in Exact@5/30, showing that memory is most useful for within-group ranking.

Model	Trainable	Group@1↑	Group@5↑	Group@10↑	Group@30↑	Exact@5↑	Exact@30↑	MAP↑
direct_clip	none	0.6335	0.8778	0.9084	0.9588	0.0012	0.0028	0.0013
head_only	head	0.7933	0.8619	0.8780	0.9406	0.0296	0.1263	0.0282
text_lora	head+T	0.7623	0.8661	0.8880	0.9420	0.0312	0.1190	0.0280
image_lora	head+V	0.7063	0.8812	0.9044	0.9550	0.0376	0.1188	0.0302
dual_lora	head+T+V	0.6275	0.8816	0.9064	0.9540	0.0326	0.1315	0.0286

Table 2. **Experiment 2: modality-SFT results on rebuild_medium.** The north-star metrics are Group@5 and Group@10, matching a chat UI that displays a small panel of sticker candidates. Exact metrics are stricter single-positive diagnostics. **Red bold** marks the best method per column and **blue bold** marks the runner-up. T/V denote text/visual LoRA.

Experiment 2: visual and dual SFT answer different needs. Once memory is fixed to `retrieved_topk`, the SFT question is which OpenCLIP branch to adapt. For a chat panel, semantic Group@5/10 should be the north star because users see multiple plausible stickers. Under that criterion, `dual_lora` is the best trained model, with Group@5 = 0.8816 and trained-model Group@10 = 0.9064. For stricter within-group ranking, `image_lora` is strongest: it has the best Exact@5 and MAP, suggesting that adapting the visual branch is especially useful for separating visually similar stickers and animated variants. Thus dual LoRA is the better first-stage retriever, while image LoRA is the strongest reference for exact top-5 ranking.

Why modalities behave differently. The per-media table shows that GIF is the most responsive format to visual adaptation: `image_lora` wins GIF Exact@5 and Group@5, while `dual_lora` wins GIF Exact@30 and Group@10. Animated stickers encode much of their intent in character pose, motion, and style, so adapting the visual branch makes the embedding space more sticker-specific. PNG behaves differently. The frozen CLIP space already separates many static stickers well, so `head_only` remains strong on PNG Exact@30, while `dual_lora` gives the best PNG Group@5/10. WEBM remains the hardest exact-retrieval case; frame sampling and video compression likely erase cues needed to distinguish close variants.

Recommended retrieval design. The two experiments suggest a two-stage design. First, keep retrieved memory because it materially improves exact selection over a no-memory head. Second, use `dual_lora` as the group-aware first-stage retriever and optimize around Group@5/10. Exact@5 and MAP should be tracked as secondary ranking signals, where `image_lora` currently sets the strongest reference point. A future reranker can combine these strengths: use `dual_lora` to guarantee semantic coverage, then apply a lightweight visual/textual reranker or diversity

rule to choose the final 5–10 stickers.

6. Conclusion

We presented MultiSticker, an intent-guided dialogue-aware sticker retrieval system that combines OpenCLIP, retrieved long-term memory, and LLM-generated intent into a trainable retrieval head with optional LoRA adaptation. The report is organized as two experiments. The memory experiment shows that a strong frozen CLIP baseline already finds semantic neighbors, but retrieved dialogue memory is what substantially improves exact sticker-id selection. The modality-SFT experiment shows that dual LoRA is the strongest trained model for group-level panel recommendation, while image-side LoRA gives the best exact top-5 ordering. Together, these results support a two-stage view of sticker retrieval: keep memory for fine-grained selection, optimize the first-stage retriever for semantically correct 5–10 item panels, and use exact metrics as diagnostics for reranking.

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Media	Model	Count	Group@5↑	Group@10↑	Exact@5↑	Exact@30↑
PNG	head_only	1,676	0.9141	0.9242	0.0197	0.1062
PNG	text_lora	1,676	0.9206	0.9314	0.0286	0.0990
PNG	image_lora	1,676	0.9254	0.9391	0.0185	0.0752
PNG	dual_lora	1,676	0.9320	0.9403	0.0304	0.0973
GIF	head_only	2,284	0.8695	0.8870	0.0499	0.1935
GIF	text_lora	2,284	0.8717	0.8971	0.0468	0.1843
GIF	image_lora	2,284	0.8927	0.9168	0.0670	0.1948
GIF	dual_lora	2,284	0.8892	0.9177	0.0464	0.2062
WEBM	head_only	1,038	0.7611	0.7832	0.0010	0.0106
WEBM	text_lora	1,038	0.7659	0.7977	0.0010	0.0077
WEBM	image_lora	1,038	0.7842	0.8208	0.0039	0.0222
WEBM	dual_lora	1,038	0.7832	0.8266	0.0058	0.0222

Table 3. **Per-media breakdown for trainable models.** Direct CLIP is omitted because its result artifact does not store per-media metrics. The group metrics are strongest and most stable in PNG/GIF, while WEBM remains the hardest exact-retrieval format.

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- Case 09 — Game, neutral_acknowledgment, gold rank 3.
- Case 10 — Game, frustration_disapproval, gold rank 4.
- Case 14 — Game, frustration_disapproval, gold rank 9.
- Case 15 — Game, frustration_disapproval, gold rank 11 (near-miss).
- Case 26 — Game, frustration_disapproval, gold rank 19 (failure).
- Case 27 — Game, neutral_acknowledgment, gold rank 20 (failure).

The PDFs live under `Latex_report/demo_assets/dual_lora_png_demo/<case>/` in the source release.

A. Additional qualitative cases

Seven additional rendered demos (chat bubble + retrieved top-5 panel + session memory + intent) accompany the case shown in Figure 3. They span a success→near-miss→failure gradient across the Game and Media-Sharing domains and the humor_teasing, neutral_acknowledgment, and frustration_disapproval intent groups:

- Case 01 — Media Sharing, humor_teasing, gold rank 1 (success).